

ULTRAVIOLET RADIATION CARCINOGENESIS

AT A GLANCE

- Ultraviolet radiation (UVR) from the sun is the most prevalent carcinogen in humans, particularly among Caucasians.
- Annual increases continue in the incidence of all types of skin cancer.
- The importance of exposure timing, dose rate, and total UVR exposure differ between the types of skin cancer.
- Sunlight leaves characteristic mutations that identify key molecular and cellular steps in skin tumor development.
- DNA repair, apoptosis, and cell signaling pathways are critical to preventing skin cancer, and UVR affects each of these processes.
- Genetic polymorphisms in genes for DNA repair, melanin, and free radical scavenging affect skin cancer risk.
- Novel, pathway-based approaches to prevent or treat skin cancer are being developed.
- Educational and behavioral approaches are key to minimizing skin cancer rates in the future.

A non-CSD (chronic sun-damaged) route involves intermittent sun exposure of sites such as the trunk. Non-CSD melanomas carry mutations in the **BRAF** or **NRAS** oncogenes—upstream regulators of cell cycle genes—and the patients often have variant alleles of the **melanocortin 1 receptor**. Intermittently exposed body sites are the locations of the rising melanoma incidence in recent decades, melanomas in patients younger than age 50 years, and increased melanoma rates near the equator. Recreational sunburn may explain these trends and the twofold higher melanoma incidence in office workers compared to outdoor workers.

Sunlight is also implicated by the susceptible population: both melanoma and non-melanoma skin cancers are more frequent in light-skinned individuals with blonde or red hair who burn rather than tan. Compared to black-skinned individuals, non-melanoma skin cancer risk increases approximately 10-fold in

Asians and 100-fold in Caucasians, with an additional 2- to 12-fold increased risk associated with blonde or, especially, red hair. The disparity is less pronounced for melanoma, with a risk ratio of approximately 1:1:15. In black skin, basal cell carcinoma (BCC) is rare even in patients with hereditary nevoid BCC syndrome (NBCCS or Gorlin syndrome). Skin tumors in black patients are often scar-related, though these may also be associated with sunlight.

Melanin-related effects contribute to skin cancer risk through several mechanisms: reduced total melanin in light skin, less protective capacity of pheomelanin compared to eumelanin, and greater production of photosensitized reactive oxygen species by pheomelanin.

Molecular epidemiology provides the most direct evidence for ultraviolet radiation (UVR) as the carcinogenic component of sunlight: **UVB signature mutations** are found in human BCC, SCC, and actinic keratosis (AK). Mutant cells are typically located in areas with elastotic dermis, indicating chronic sun exposure.

UVB is particularly effective as a carcinogen because it partially penetrates the ozone layer and the stratum corneum and is readily absorbed by DNA. The ozone layer absorbs all but 1 part per million of UVC (which is lethal and used in germicidal lamps), while UVA penetrates deeply but is poorly absorbed by DNA. Nevertheless, chronic UVA exposure can induce tumors in mice and can malignantly transform predisposed human cells.

The cumulative sunlight exposure required to cause BCC or SCC in adults is substantial—approximately 10,000 hours for BCC and 70,000 hours for SCC. Psoriasis patients receiving fewer than 100 UVB treatments (with low psoralen plus UVA [PUVA] exposure) had a slightly elevated risk of SCC or BCC on sun-shielded sites, while those receiving more than 300 UVB treatments had a threefold higher risk.